

Structuring the Unstructured: An NLP-Driven Clinical Data PoC for Tempus AI in Ireland

Executive Summary

This report proposes an AI project plan for Tempus AI, building on the Group Assignment (group 14), which identified a key bottleneck: although the company holds 40 million patient records, only 22% are AI-ready (Tempus AI, 2025b). The group strategy introduced an Agentic NLP Abstraction Engine to address this issue.

This paper develops a focused Proof-of-Concept (PoC): an NLP pipeline that extracts and structures Type 2 Diabetes Mellitus (T2DM) data from GP free-text records. Ireland is selected as the test environment due to its underdeveloped health data infrastructure, making it both a strategic market and a realistic GDPR-constrained deployment context. China's regional platforms demonstrate technical feasibility, but require adaptation to the Irish setting.

The solution integrates ClinicalBERT for Named Entity Recognition, OMOP CDM v5.4 for data standardisation, Microsoft Presidio for PHI de-identification, and human-in-the-loop validation. The 12-week PoC targets 500,000 GP records, aiming to increase structured data coverage from around 18% to over 45%.

1. The Problem: A Library No One Can Read

Tempus AI faces a structural data bottleneck. Despite operating at scale across genomics and clinical AI, only 22% of its data is structured and usable, while 78% remains in free text (Tempus AI, 2025b). This limits query ability and reduces the commercial value of its datasets.

GP records are particularly important, as they provide longitudinal insight into chronic disease management, including medication, lab results, and care pathways. For T2DM, they represent the most complete patient history. Structuring this data directly supports Tempus's Insights business, which relies on high-quality longitudinal datasets.

1.1 What China Has Already Demonstrated

China's regional health information platforms (e.g. Ningbo, Guangdong) have already achieved large-scale integration of primary care data, including structured extraction from free-text records (Li et al., 2020). For example, Ningbo integrated thousands of facilities and structured chronic disease data such as T2DM lab results and medication histories within months. This demonstrates that large-scale structuring of GP-like data is technically feasible. However, these Chinese platforms were government-mandated public health infrastructure projects. Participation was compulsory, funding was city-provided, and there was no requirement to satisfy GDPR, commercial licensing obligations, or voluntary GP consent. The analogy used here is strictly limited to the technical feasibility that NLP structuring of primary care records at scale is achievable. The commercial model, governance framework, and legal constraints for this PoC are entirely different.

1.2 Why Ireland Specifically

Ireland's health data infrastructure is remarkably underdeveloped by European standards, which creates both a challenge and an opportunity. As of 2024 (data as of 31 December 2024), Ireland's eHealth maturity score stands at 25% (the lowest in the EU) against an EU-27 average of 83% (European Commission, 2025a). Until 2025, Ireland was one of the EU member states without any online portal

through which citizens could access their own health records (Euronews, 2026). The 2021 ransomware attack on the HSE, which was the most severe cyberattack on any national health service that year, compounded this fragility and raised security sensitivity throughout Irish healthcare (Health Service Executive, 2025). In February 2026, the government approved procurement for a National Electronic Health Record, the first concrete step towards a unified system (Euronews, 2026). The Health Information Bill 2024 creates a statutory duty to share health information (Department of Health, 2024). This is the regulatory moment in which a Tempus GP NLP PoC becomes both viable and timely.

1.3 Why China's approach cannot be directly transposed

Three constraints make Ireland fundamentally different. Firstly, GDPR Article 17 right to erasure requires all derived structured records to be deletable on patient request that a Delta Lake DELETE workflow that must be designed before go-live. Secondly, voluntary GP participation means health systems must be convinced, not mandated. Thirdly, Irish-English GP shorthand ("T2DM x14 yrs", "HbA1c 58 - stable", "review 3/12") is entirely different from the Mandarin clinical text on which Chinese NLP models were trained. These show that the technical precedent transfers but the design does not.

2. The Approach: Three Everyday Analogies

The core technical method is clinical Named Entity Recognition (NER) combined with ontology mapping. Three analogies make the key components accessible without sacrificing precision.

Analogy	Technical Component	How It Works in This PoC
The Parcel Sorter	NER - ClinicalBERT (Alsentzer et al., 2019)	A parcel sorter reads an address and classifies each span (person, street, postcode) without understanding what it means. ClinicalBERT does the same with GP free-text: <i>"Patient started metformin 500 BD, HbA1c now 58 - stable, no nephropathy noted"</i> → metformin 500 BD = medication entity; HbA1c 58 = lab value; nephropathy = negated diagnosis (via NegEx). "No nephropathy" must not be recorded as a nephropathy diagnosis.
The Currency Exchange	OMOP CDM v5.4 (OHDSI, 2023)	Dublin GP writes <i>"metformin 500 BD"</i>; Manchester GP writes <i>"Metformin 500mg twice daily"</i>; Boston clinician writes <i>"Metformin 500 BID"</i>. Three strings, one clinical meaning. OMOP CDM maps all three to RxNorm concept 860975, making cross-site and cross-acquisition queries possible across the Tempus ecosystem.
Airport Security Queue	Confidence Scoring + Active Learning (Settles, 2009)	Most bags pass automatically; ambiguous ones go to human inspection; a small number are detained. Human judgement is applied only where it adds value. Extractions with confidence ≥ 0.85 are auto-promoted to the Silver layer. Those below 0.85 enter a human review queue. Validated corrections feed back into monthly ClinicalBERT retraining cycles (active learning loop).

3. Architecture and Pipeline

This PoC operationalises Layers 1-4 of the Group Assignment’s 6-layer Unified Clinical AI Platform (Bronze ingestion, Silver harmonisation, Agentic NLP Engine, Gold serving). No separate infrastructure is proposed; the Medallion Lakehouse on Azure Data Lake Storage Gen2 with Delta Lake is adopted without modification.



Figure 1 Seven-stage GP NLP structuring pipeline

Stage 1 (Bronze) ingests GP notes via Apache Kafka on Azure Event Hubs, stored as append-only, immutable Delta Lake tables. Stage 2 (PHI De-identification) applies Microsoft Presidio in-path (<50ms), supplemented by Azure AI Language Healthcare API. Only de-identified text proceeds, satisfying both GDPR Article 5 data minimisation and the HIPAA audit retention requirement. Stage 3 (NER) deploys ClinicalBERT on Databricks ML Runtime to extract six entity types: diagnoses (SNOMED CT), medications (RxNorm), lab values (LOINC), dosage/frequency, temporal markers, and clinical status flags. Relative temporal expressions (“started insulin 2 winters ago”) are normalised by SUTime (Chang and Manning, 2012). Stage 4 (Confidence Scoring) routes extractions below 0.85 to a Silver Review Queue; corrections trigger active learning retraining. Stage 5 (OMOP Harmonisation) writes validated extractions to the Silver layer using the OHDSI Athena vocabulary service; patient entity resolution links GP records to existing Tempus identifiers. Stages 6-7 publish to Gold marts for pharmaceutical Insights licensing, TIME trial matching, and FUSES Foundation Model training, with every extraction event logged to immutable Delta Lake audit tables via Microsoft Purview.

Component	Specification	Justification
Compute	Databricks D16s v3 (16 vCPU, 64GB)	ClinicalBERT ~40,000 records/hour; auto-scalable
PHI Scanning	Microsoft Presidio (<50ms in-path)	GDPR-compliant within Azure EU West boundary
Model Registry	MLflow + Unity Catalog	Version control, A/B testing, 5-min RTO rollback
Vector Indexing	Azure AI Search	Semantic search on structured output for Tempus One

Stage 2 Deployment about Ownership, IP, and Scope: Successful completion of this PoC triggers a formal go/no-go decision before any Stage 2 deployment. Three governance questions must be resolved at that decision point.

Governance Question		Stage 2 Requirement
1. Data Ownership	Who owns the structured records?	Silver and Gold layer records derived from GP free-text remain subject to the originating health system's Data Processing Agreement. Tempus holds a licensed right to use, not an unencumbered ownership claim.
2. Intellectual Property	Who owns the model and annotations?	The fine-tuned ClinicalBERT weights trained on annotated Irish GP records are a proprietary Tempus asset. However, the underlying annotations require explicit IP assignment clauses in contracts with GP-trained annotators before training begins.
3. Scope Boundary	What triggers Stage 2 expansion?	Stage 2 (full 31M-record backlog; cardiovascular disease, COPD) proceeds only after: (a) T2DM PoC Success Gate is cleared (b) commercial terms confirmed with ≥ 1 HSE-affiliated health system. Extracting non-T2DM entities "for future use" during the PoC window violates GDPR Art. 5 data minimisation and is expressly excluded.

4. Ireland-Specific Design Challenges

Three GDPR requirements materially affect PoC design beyond what the Group Assignment's HIPAA-oriented architecture addressed:

Lawful basis (Art. 6 & 9): Data Processing Agreements with each originating health system, and potentially research ethics approval, are required before any real patient data enters the pipeline.

Clinical NLP for commercial AI training is not automatically covered by the clinical care exemption.

Right to erasure (Art. 17): If a patient exercises the right to erasure, all derived structured Silver and Gold layer records must be deleted which includes mart entries. Delta Lake DELETE operations with Time Travel confirmation must be built before go-live, not retrospectively.

Data minimisation (Art. 5): Only the defined T2DM entity types may be extracted; full record text must not be retained post-extraction. Extracting all clinical entities from a record "for future use" would not satisfy this principle.

GDPR Requirement	PoC Implementation
Lawful basis (Art. 6 & 9)	Data Processing Agreements; research ethics approval for annotation
Data minimisation (Art. 5)	Extraction limited to T2DM entity types; full text deleted post-extraction
Right to erasure (Art. 17)	Delta Lake DELETE; Time Travel confirmation; operational process pre-designed
Cross-border transfers	All processing in Azure EU West; Microsoft Azure DPA + SCCs verified
Audit trail (Art. 30)	Immutable Delta Lake logs; Microsoft Purview cataloguing

5. Model Selection and Evaluation

5.1 Why ClinicalBERT, Not GPT-4

A generative LLM approach was considered and rejected for three reasons. First, pharmaceutical audit requirements: Tempus's partners require exact reproducibility linking every extracted entity to its source span, model version, and confidence score. However, generative models provide neither extraction-level confidence scores nor deterministic reproducibility. Second, GDPR compliance requires that sending patient data to an external API requires cross-border transfer controls substantially more complex than operating within the Azure tenant boundary. Third, API-based inference at 500,000 records is an order of magnitude more expensive than a fine-tuned BERT-class model on Databricks.

ClinicalBERT is pre-trained on MIMIC-III clinical notes (Johnson et al., 2016), substantially outperforms general BERT on clinical NER including medication extraction and lab value recognition, and produces per-entity confidence scores essential for the routing logic. It is available on Hugging Face and versioned via MLflow.

ClinicalBERT was trained on hospital discharge summaries that are longer, more structured, and written by specialist clinicians. GP notes are shorter, more colloquial, and abbreviation-dense. If entity-level F1 on the annotated test set falls below 0.78, the PoC should be paused for a GP-specific fine-tuning sprint before proceeding.

5.2 Evaluation Targets

Metric	Target	Method
Entity-level F1 (overall)	>0.82	Held-out annotated test set (5,000 records)
Entity-level F1 (medications)	>0.88	RxNorm-validated medication subset
Entity-level F1 (lab values)	>0.90	LOINC-validated lab value subset
PHI leakage rate	0%	Independent red-team audit with synthetic PHI injection
Structured data coverage	>45%	Records promoted to Silver with confidence ≥ 0.85
Human review queue rate	<15%	Records requiring manual review per batch
OMOP mapping success rate	>92%	Athena vocabulary match rate on extracted entities

6. Team, Roadmap, and Anticipated Challenges

6.1 Required Personnel (~6 FTE)

Role	FTE	Responsibilities
NLP / ML Engineer	2	ClinicalBERT fine-tuning, pipeline development, MLflow integration
Clinical Annotator (GP-trained)	2 pt	Ground truth annotation (5,000 records); IAA ≥ 0.80
Data Engineer	1	Bronze ingestion, OMOP mapping, Silver harmonisation
Data Governance Lead	1	GDPR compliance, Unity Catalog RBAC, audit trail management
Clinical Informatics Lead	1	OMOP mapping review, human review queue management

6.2 12-Week Roadmap

Phase	Weeks	Key Deliverables
1.Foundation	1-3	Bronze pipeline operational PHI de-id validated ClinicalBERT deployed DPAs signed Annotation schema defined
2.Extraction	4-7	NER on 100K pilot corpus 5,000 records annotated (IAA ≥ 0.80) F1 evaluation Review queue live
3.Harmonisation	8-10	OMOP mapping complete Gold mart published Full 500K run Evaluation report Fairness analysis
4.Governance	11-12	Model Observatory validated GDPR documentation complete Go/no-go decision

PoC Success Gate: Bronze-to-Silver with zero PHI leakage, NER F1 > 0.82, Structured coverage > 45%, GDPR compliance documented and Model Observatory audit trail are validated.

6.3 Formal Go/No-Go Decision

At the conclusion of Week 12, the project sponsor and data governance lead conduct a structured go/no-go review against the five Success Gate criteria above. This is a binary decision gate, not a sliding scale. All five criteria must be met to proceed to Stage 2 deployment.

No-go path: If one or more criteria are not met, three outcomes are possible. If $F1 < 0.82$ but PHI leakage = 0% and GDPR is compliant, the project enters a defined GP-specific fine-tuning sprint (maximum 6 weeks, capped budget of €30K) before a second gate review. If PHI leakage > 0% at any point during the PoC, the pipeline is suspended immediately regardless of other metrics; no further processing occurs until a root cause analysis is completed and an independent red-team audit passes. If GDPR documentation is incomplete at Week 12, Stage 2 cannot proceed under any circumstances; this is a hard legal constraint, not a performance threshold.

If the problem is not modellable: A distinct failure mode exists beyond underperformance. The possibility that Irish GP free-text is structurally resistant to ClinicalBERT-class NER at the required accuracy threshold, even after fine-tuning. If a GP-specific fine-tuning sprint fails to reach $F1 = 0.82$, the project team must assess whether the failure reflects a data quality problem (resolvable with more annotations), a model architecture problem (requiring a different model class), or a fundamental domain mismatch (Irish GP shorthand is too idiosyncratic for current NLP methods). In the last case, the PoC result is itself valuable. It establishes an evidence-based ceiling on current NLP capability for this data type and Tempus should document the finding, preserve the annotated test set as a future benchmark, and escalate to the FUSES Foundation Model team for a longer-horizon solution. Failure to model the problem does not invalidate the business case. It redirects the technical pathway.

6.4 Anticipated Challenges

Ground truth scarcity is that 5,000 annotated GP records require GP-trained clinicians and ICD billing codes as proxy labels introduce noise. An inter-annotator agreement of ≥ 0.80 (Cohen's Kappa) is required before use as a gold standard.

Review queue burden is that if $>20\%$ of records fall below 0.85 threshold, reviewer fatigue and backlog accumulate. Monthly per-entity precision reporting from week one identifies underperforming entity types early.

Algorithmic fairness is that T2DM documentation patterns may vary across demographic groups in Ireland's increasingly diverse population. Subgroup F1 stratification by age, sex, and socioeconomic status should be conducted using the annotated test set as part of the evaluation framework. Subgroup F1 analysis is required. The Group Assignment's 5% F1 gap threshold could apply as a deployment gate.

The scale boundary is that this PoC validates the approach on a controlled corpus. Clearing the full 31-million record backlog is the enterprise pipeline objective. And the D16s v3 cluster is appropriate for the PoC scope only.

7. Financial Case: The Cost of Inaction and the Return on Investment

The business case for this PoC cannot rest solely on technical merit. It must demonstrate that the status quo (Ireland's current fragmented, unstructured GP data landscape) carries a quantifiable and growing financial cost, and that the investment in NLP-driven structuring produces a return that exceeds that cost within a commercially credible timeframe. This section establishes both sides of that equation using publicly available Irish healthcare data.

7.1 The Cost of the Status Quo: Ireland's T2DM Burden Without Structured Data

Type 2 Diabetes Mellitus is Ireland's most expensive manageable chronic condition. In 2022, diabetes featured as a principal or additional diagnosis in 8.6% of all coded hospital admissions in Ireland (146,362 hospitalisations out of 1,711,564 total) at a total inpatient cost of €601 million (O'Sullivan et al., 2024). Of those admissions, 58.5% were T2DM cases, implying a T2DM-attributable inpatient cost of approximately € 351 million in a single year.

At the patient level, the direct annual cost of T2DM in Ireland was estimated at €2,469 per patient in 2006, with €1,196 attributable to hospitalisation alone, because 60% of T2DM patients develop diabetes-related complications that require inpatient treatment (Nolan et al., 2006). Applying cumulative Irish CPI inflation of approximately 2.4× from 2006 to 2024 (CSO Ireland), these figures translate to approximately €5,900 per patient per year in 2024 terms, with roughly €2,900 attributable to hospitalisation. No more recent Ireland-specific per-patient T2DM cost study was identified. These inflation-adjusted estimates are used as a conservative proxy. The broader picture is consistent with more recent sources: Friel et al. (2022) confirm that the cost of managing diabetes now represents an estimated 12-14% (approximately €2 billion) of Ireland's annual health budget. Each additional hospital day adds approximately €216 in cost at 2022 prices. Patients with documented complications cost an average of €721 more per admission than those without (Friel et al., 2022). With an estimated approximately 270,000 people living with T2DM in Ireland (HSE, 2024) and the data is derived from

Scottish prevalence data applied to the Irish population and no national registry exists, and the cost of complications rising as the population ages, the aggregate burden is substantial and growing.

Ireland currently has no national diabetes registry and no structured electronic diabetes record. As Diabetes Ireland (2023) states: ‘with no data - no diabetes registry nor clinical audits - we cannot even fully identify what the gaps and issues are.’ This is precisely the problem this PoC addresses. Without structured GP records, the HSE cannot identify which patients are poorly controlled, cannot target preventive interventions, and cannot measure whether care improvements are working.

The financial implication is stark. Research consistently shows that complications (nephropathy, retinopathy, neuropathy, cardiovascular events) are the primary driver of diabetes costs, and that many of these complications are preventable through earlier detection and more targeted management (Nolan et al., 2006; Diabetes Ireland, 2023). A 10% reduction in avoidable T2DM complication-related hospitalisations in Ireland would represent a saving of approximately €35 million per year at current cost levels. The NLP pipeline proposed in this PoC creates the structured data layer that makes population-level identification and targeting of high-risk patients possible for the first time.

Metric	Value	Source
T2DM inpatient admissions, Ireland 2022 (est.)	~85,600	O'Sullivan et al. (2024), HIPE
Total diabetes inpatient cost, Ireland 2022	€601 million	O'Sullivan et al. (2024)
Estimated T2DM share (58.5%)	~€351 million/year	Calculated from HIPE 2022 split
Average annual direct cost per T2DM patient	€2,469 (2006); ~€5,900 inflation-adjusted (2024) [CPI-adjusted]	Nolan et al. (2006), Irish Medical Journal
Average hospitalisation cost per T2DM patient/yr	€1,196 (2006); ~€2,900 inflation-adjusted (2024) [CPI-adjusted]	Nolan et al. (2006)
Estimated T2DM patients in Ireland	~270,000 (est.)	HSE National Clinical Programme (2024)
Marginal cost per additional hospital day (T2DM)	€216	Friel et al. (2022), Diabetic Medicine
Additional cost if complications present	+€721 per admission	Friel et al. (2022), Diabetic Medicine

Note: Ireland has no national diabetes registry; prevalence figures are derived from HSE estimates based on Scottish prevalence data applied to the Irish population; the figure of ~270,000 is an approximation, not a precise count. The 58.5% T2DM share of inpatient diabetes admissions is from 2022 HIPE data (O'Sullivan et al., 2024). Per-patient cost figures from Nolan et al. (2006) have been inflation-adjusted using an approximate 2.4× CPI multiplier (CSO Ireland, 2006-2024); the original 2006 values were €2,469 and €1,196 respectively. No more recent Ireland-specific per-patient T2DM cost study was identified.

7.2 The Cost Structure of the PoC Investment

The PoC investment is front-loaded but modest relative to the problem it addresses. The dominant cost items in Year 0 are personnel (NLP/ML engineers and clinical annotators) and Databricks compute. Infrastructure costs are minimal because the PoC is built on top of the existing Medallion Lakehouse architecture designed in the Group Assignment, which means no new cloud storage, no new security layer, and no new orchestration tooling is required.

From Year 1 onwards, costs grow as the pipeline scales to enterprise throughput, but revenue from Tempus's Insights licensing division begins to offset operational expenditure. The revenue projections below are conservative, based on Tempus's publicly reported 40.7% year-on-year growth in Insights licensing revenue in Q2 2025 (Tempus AI, 2025b) and structured T2DM cohort data commanding a material premium over unstructured records in pharmaceutical licensing markets.

Cost / Revenue Item	Year 0 (PoC)	Year 1	Years 2-3 (p.a.)	Years 4-5 (p.a.)
COSTS (investment & operating)				
NLP/ML Engineers (2 FTE, 12 wks.)	-€120K	-€60K	-€80K	-€80K
Clinical Annotators (2 × part-time)	-€40K	-€20K	-€10K	-€10K
Databricks D16s v3 compute	-€25K	-€60K	-€80K	-€80K
Legal / DPA / GDPR compliance	-€15K	-€10K	-€5K	-€5K
TOTAL COSTS	-€200K	-€150K	-€175K	-€175K
REVENUE (Tempus commercial upside)				
Insights licensing - T2DM cohorts	-	+€300K	+€600K	+€900K
TIME trial matching uplift	-	+€50K	+€200K	+€350K
FUSES Foundation Model training value	-	+€20K	+€80K	+€150K
TOTAL REVENUE	-	+€370K	+€880K	+€1,400K
NET POSITION				
Net cashflow (period)	-€200K	+€220K	+€705K	+€1,225K
Cumulative net (from Year 0)	-€200K	+€20K	+€1,430K	+€3,880K

All figures are illustrative projections for planning purposes, not audited forecasts. Revenue estimates are based on Tempus's publicly reported Insights licensing growth trajectory (Tempus AI, 2025b) and conservative assumptions about cohort pricing. Compute costs based on Databricks D16s v3 at approximately €0.15/DBU-hour for PoC scope, scaling to D64s for enterprise rollout. Legal and GDPR compliance costs reflect typical Data Processing Agreement and research ethics review timelines in the Irish healthcare context.

7.3 Return on Investment and Break-Even Analysis

Based on the projections above, the cumulative net position turns positive during Year 1, with full break-even on the PoC investment of €200K achieved within approximately 12 months of go-live. From Year 2 onwards, the structured GP data asset generates net annual returns of approximately €700,000, rising to over €1.2 million per year in Years 4-5 as the pipeline expands beyond T2DM to additional chronic conditions (cardiovascular disease, COPD) and FUSES Foundation Model training data becomes a material commercial asset.

The 5-year cumulative net benefit to Tempus is approximately €3.9 million. It is a 19.5x return on the initial PoC investment. This figure does not capture the broader value accruing to the Irish healthcare system from improved T2DM management. It should be noted critically that the revenue projections underpinning this return (particularly the Year 1 Insights licensing figure of €300K) are illustrative scenarios, instead of evidence-based forecasts. They are not derived from any confirmed contract or market rate for Irish GP T2DM cohort data. The 40.7% Tempus Insights growth rate cited reflects company-wide performance and cannot be extrapolated to a new regional PoC. Readers should treat the ROI figures as order-of-magnitude estimates to justify initiating the PoC, subject to commercial validation. Furthermore, the ClinicalBERT fine-tuning sprint which is a realistic probability given the domain shift from MIMIC-III to Irish GP text and would add cost and delay not reflected in the current projections.

ROI metric	Estimate	Basis
PoC upfront investment	€200,000	Personnel, compute, legal (Year 0)
Year 1 net position	+€220,000	First Insights licensing revenue offset against scale-up costs
Break-even from go-live	~12 months	Cumulative cashflow turns positive in Year 1
5-year cumulative net benefit	~€3.9 million	Revenue from structured GP data assets, net of all costs
ROI on PoC investment (5-year)	~19.5x	5-year net / PoC upfront cost
HSE complication cost savings (10% reduction scenario)	~€35M/year	10% of €351M T2DM inpatient cost; attributable to improved early detection

The HSE complication cost saving is a societal externality, not a direct Tempus revenue line, and is therefore excluded from the Tempus commercial ROI calculations above. It is presented here as a separate social value metric, not as part of the 19.5x ROI figure. Conflating this externality with Tempus's commercial return would misrepresent the business case. Its relevance is to demonstrate system-level value to HSE and pharmaceutical partners - not to inflate Tempus's own return on investment.

7.4 Why the Investment Is De-Risked

Three features reduce the financial risk of this PoC relative to a typical greenfield AI project.

The pipeline builds on the existing Medallion Lakehouse architecture developed in the Group Assignment, limiting incremental costs to compute scaling and pipeline-specific tools rather than a full infrastructure build.

A 12-week PoC tests performance on 500,000 records before any scale-up. A go/no-go decision at Week 12 ensures that failure to meet the F1 threshold of 0.82 prevents further investment.

Tempus's Insights division already serves 70+ pharmaceutical clients with 126% net revenue retention (Tempus AI, 2026a; Tempus AI, 2026b). T2DM structured datasets extend an established, revenue-generating product line.

At a system level, Ireland incurs approximately €351 million annually in T2DM inpatient costs, much of it linked to preventable complications. The €200,000 PoC reaches break-even within 12 months and generates an estimated €3.9 million return over five years, while also enabling downstream cost reductions for the HSE and pharmaceutical partners through improved data availability.

7.5 Stage 3 Maintenance: Long-Run Governance Framework

A deployed AI system is not a finished product. Stage 3 Maintenance begins at the moment Stage 2 go-live is approved and governs the operational life of the structured GP data pipeline. Nine governance obligations apply from day one of production.

Governance Obligation	Requirement (Stage 3 Production)
1. Explainability (XAI)	Every extraction must link to its source GP text span and SHAP confidence score. Black-box outputs are not permitted in the Silver review queue.
2. Bias & Fairness	Monthly subgroup F1 by age, sex, SES via Model Observatory. A >5 pp gap vs. overall population triggers mandatory retraining before the next batch.
3. Liability Allocation	Tempus liability bounded to structured data accuracy per DPA. Clinical decisions remain the clinician's responsibility. EU AI Act high-risk classification requires auditable liability chain before production.
4. Data Lineage	Every Gold record carries: source system, ingestion timestamp, model version, confidence score, reviewer ID, OMOP version, Delta Lake commit hash. Enforced by Microsoft Purview (GDPR Art. 30).
5. Autonomy / Opt-Out	Patient opt-out triggers Art. 17 erasure of all Silver/Gold records plus downstream licensee notification. Opt-out friction must not exceed opt-in friction.
6. Model Versioning & Rollback	MLflow versioning; prior version retained 90 days. F1 regression >3 pp over 30 days triggers 5-min rollback (Group Assignment RTO). New versions must pass Success Gate F1 on held-out validation before production.
7. Data & Concept Drift	Weekly Population Stability Index on entity distribution. Detected drift triggers targeted annotation sprint before production throughput is affected.
8. Third-Party Audit Rights	HSE and Data Protection Commission may audit annually or on complaint. Scope: Model Observatory logs, Delta Lake lineage, Purview catalogue, annotated test set. Required under GDPR Art. 28; must appear in every DPA.
9. Sunset & Decommissioning	On programme termination: Bronze raw text deleted within 30 days; Silver/Gold returned to health system or deleted; model weights retained 5 years for pharma regulatory audit. Decommissioning is a contractual obligation, not a discretionary act.

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